AI-Based Statistics Tutor Project

1. Introduction

This project presents an AI-Based Statistics Tutor developed using the Phi-3-mini-4k-instruct language model. The system answers statistics-related conceptual and educational questions through conversational interaction.

The project uses RunPod cloud GPU services for deployment and vLLM as the inference engine. Python was used for API integration and user interaction.

The main objective of the project was to create a simple AI tutor capable of helping students understand statistics concepts.

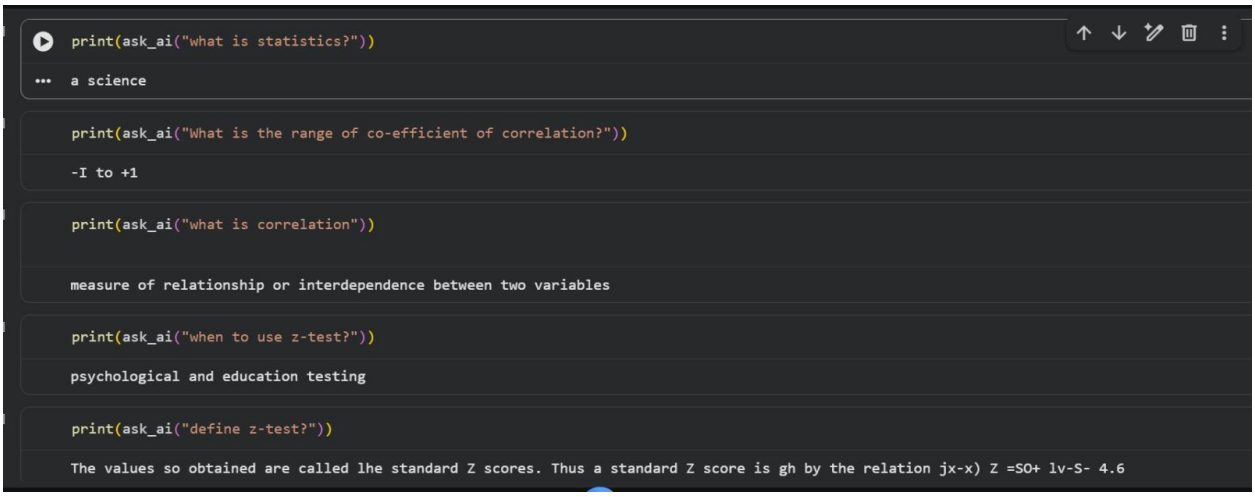
2. Dataset / Initial Approach

Initially, the project was developed using PDF-based Retrieval-Augmented Generation (RAG), where statistics PDFs were used as the knowledge source.

However, the system did not produce satisfactory responses because of:

- Poor retrieval quality
- Incomplete contextual understanding
- Noisy PDF extraction
- Irrelevant answers in some cases

Therefore, the final system was redesigned using prompt engineering with the pretrained Phi-3-mini model instead of PDF retrieval.



3. Methodology

The proposed AI tutor system was developed using the Phi-3-mini-4k-instruct language model with API-based interaction.
The workflow of the proposed system is as follows:

1. User enters a statistics-related question.
2. Python client sends request to deployed API.
3. RunPod hosts the Phi-3-mini model using vLLM.
4. The model processes the prompt.
5. AI-generated response is returned to the user.
6. Conversation history is maintained using message memory.

Technologies Used

Component	*Purpose*
Phi-3-mini	Language Model
vLLM	Inference Engine
RunPod	Cloud GPU Hosting (API Server)
Python (Google Colab)	Client for sending API requests
Requests Library	Handles HTTP API communication

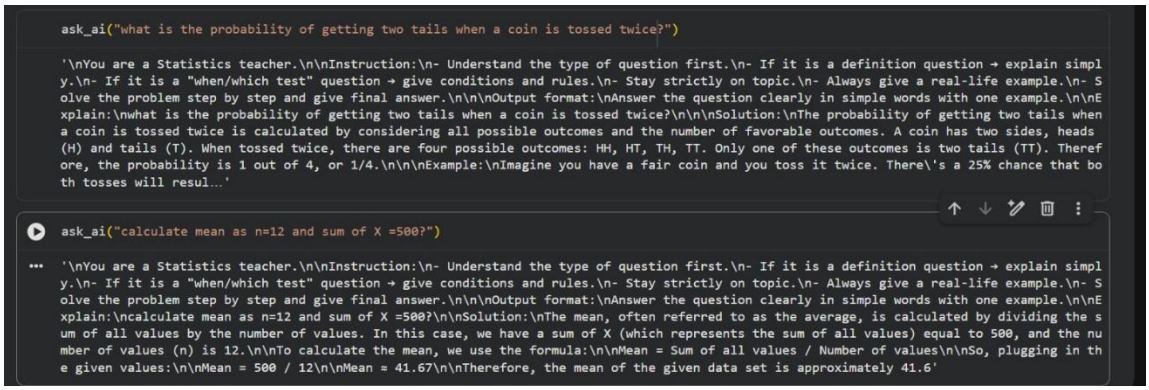
Prompt Engineering

Prompt engineering was used to guide the model behavior.

Example system prompt:

"You are a helpful statistics tutor. Give only final answers clearly."

This improved the quality and clarity of educational responses.



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ask_ai("what is the probability of getting two tails when a coin is tossed twice?")

'\nYou are a Statistics teacher.\n\nInstruction:\n- Understand the type of question first.\n- If it is a definition question → explain simpl
y.\n- If it is a "when/which test" question → give conditions and rules.\n- Stay strictly on topic.\n- Always give a real-life example.\n- S
olve the problem step by step and give final answer.\n\n\nOutput format:\nAnswer the question clearly in simple words with one example.\n\nE
xplain:\nwhat is the probability of getting two tails when a coin is tossed twice?\n\n\nSolution:\nThe probability of getting two tails when
a coin is tossed twice is calculated by considering all possible outcomes and the number of favorable outcomes. A coin has two sides, heads
(H) and tails (T). When tossed twice, there are four possible outcomes: HH, HT, TH, TT. Only one of these outcomes is two tails (TT). Theref
ore, the probability is 1 out of 4, or 1/4.\n\n\nExample:\nImagine you have a fair coin and you toss it twice. There\'s a 25% chance that bo
th tosses will resul...'

ask_ai("calculate mean as n=12 and sum of X =500?")

'\nYou are a Statistics teacher.\n\nInstruction:\n- Understand the type of question first.\n- If it is a definition question → explain simpl
y.\n- If it is a "when/which test" question → give conditions and rules.\n- Stay strictly on topic.\n- Always give a real-life example.\n- S
olve the problem step by step and give final answer.\n\n\nOutput format:\nAnswer the question clearly in simple words with one example.\n\nE
xplain:\ncalculate mean as n=12 and sum of X =500?\n\nSolution:\nThe mean, often referred to as the average, is calculated by dividing the s
um of all values by the number of values. In this case, we have a sum of X (which represents the sum of all values) equal to 500, and the nu
mber of values (n) is 12.\n\nTo calculate the mean, we use the formula:\n\nMean = Sum of all values / Number of values\n\nSo, plugging in th
e given values:\n\nMean = 500 / 12\n\nMean = 41.67\n\nTherefore, the mean of the given data set is approximately 41.6'
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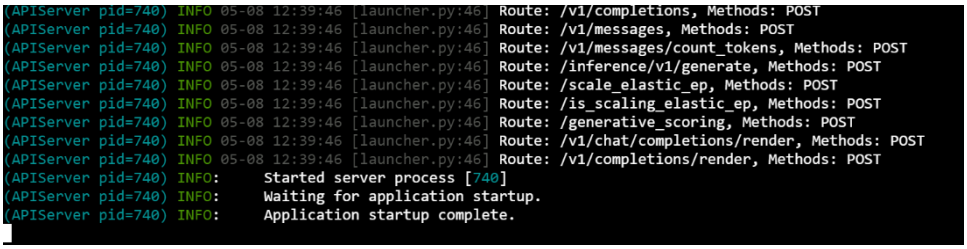
4. Deployment Steps

The deployment process was completed using RunPod and vLLM.

Steps Performed

1. Created RunPod GPU instance.
2. Installed vLLM framework.
3. Loaded Phi-3-mini-4k-instruct model.
4. Started OpenAI-compatible API server.
5. Generated API endpoint.
6. Connected Python client using requests library.
7. Tested responses through terminal interaction.



5. Evaluation

The system was evaluated using conceptual and problem-solving statistics questions.

Test Type	*Result*
Conceptual Questions	Successful
Statistics Problem Solving	Successful
Conversational Interaction	Successful
API Communication	Successful

Example Evaluation

Client Question:

When two coins are tossed, what is the probability of getting two heads?

AI Response:

The probability of getting two heads is 1/4.

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•••  You: when 2 coins are tossed what is the probability that both are tails
     AI:  The probability of getting tails on a single coin toss is 1/2.

     Since the two coin tosses are independent events, you multiply the probabilities of each event occurring. So the probability of getting tails

     (1/2) * (1/2) = 1/4 or 0.25.

     Final answer: The probability of getting tails on both coin tosses is 0.25.
     You: what is the difference between median and mean
     AI:  The median and the mean are both measures of central tendency used to describe the center of a data set, but they differ significantly i


     The median is the value at the midpoint of a data set when it is ordered from least to greatest. If there is an even number of observations,


     The mean, on the other hand, is calculated by adding up all the numbers in the data set and then dividing by the total number of observations
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6. Conclusion

This project developed an AI-Based Statistics Tutor using the Phi-3-mini model deployed on RunPod with vLLM. Initially, a RAG-based approach using PDF documents was implemented, but it produced inconsistent and low-quality responses due to weak retrieval and noisy data processing. Therefore, the system was redesigned using prompt engineering instead of document-based retrieval. The final system effectively answers conceptual and educational statistics questions through API-based conversational interaction.

However, its performance is still limited by the pretrained knowledge of the model, as no fine-tuning was performed.